Name:

Class:

ACTIVITY SHEET

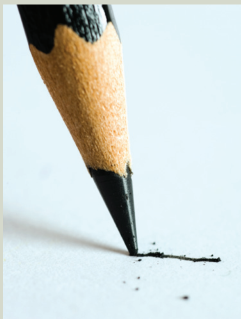
3.3 Identifying the nature of the bonding in everyday substances

The table below summarises common properties that can be used to distinguish between metallic, ionic, covalent molecular and covalent network substances.

Property	Metallic	Ionic	Covalent molecular	Covalent network
Melting and boiling points	Variable but most commonly high	High	Low	High
Electrical conductivity	Good conductor in solid and liquid states	Non-conductor in solid state but good conductor in molten and aqueous state	Does not conduct	Does not conduct (except graphite)
Hardness and malleability	Hard and malleable	Hard and brittle	Mostly soft	Hard and brittle
Solubility	Generally insoluble	Generally soluble in water	Variable – depends on solvent	Insoluble

Instructions

Identify as many properties of each substance as you can. Some have been done for you. Predict the nature of the bonding by referring to the table above.

Substance / Properties	 Shutterstock.com/PanicAttack Salt	 Shutterstock.com/hxdbzxy Graphite	 Shutterstock.com/Sergiy Kuzmin Aluminium
Melting point		Very high	
Boiling point	Not relevant	Not relevant	Not relevant
Thermal conductivity	Very poor	Good	
Electrical conductivity	Non-conductor		
Hardness			Quite hard
Malleability	Not malleable	Not malleable	
Ductility	Not ductile	Not ductile	
Solubility in water			
Type of bonding			

Substance / Properties	 Shutterstock.com/Atelier_A Sugar	 Shutterstock.com/design56 Glass	 Shutterstock.com/MNI Polythene
Melting point			
Boiling point	Not relevant	Not relevant	Not relevant
Thermal conductivity	Not relevant		Poor
Electrical conductivity	Not relevant	Insulator	
Hardness			
Malleability	Not relevant		Not relevant
Ductility	Not relevant		Not relevant
Solubility in water			
Type of bonding			

Substance / Properties	 Shutterstock.com/Viktor1 Butter	 Alamy/David J. Green Washing soda	 Shutterstock.com/idea for life Copper
Melting point		Decomposes at 50°C	
Boiling point	Not relevant	Not relevant	Not relevant
Thermal conductivity	Not relevant	Not relevant	
Electrical conductivity	Not relevant	Not relevant	
Hardness			
Malleability	Not relevant	Not relevant	
Ductility	Not relevant	Not relevant	
Solubility in water			
Type of bonding			